

Release Notes

for BLN_S32K344_SBAF_HSEB_RTM_0.15.0

Rev. 1.0 — 28 August 2023

Release notes
COMPANY CONFIDENTIAL

1 Getting Started

1.1 Package content

This package contains the NXP S32K344 SBAF 0.15.0:

- FULL MEM SBAF: encrypted binary
- The license.txt file and the `uninstall.exe` utility for removing the SBAF binaries



2 Release Details

This is the HSE SBAF 0.15.0 RTM release for S32K344 and is ready for production. It is strongly recommended that user should update the SBAF in their existing samples if the features provided in this release are important to them.

This release was developed and tested using:

- Chip : E5 - P32K344EHVMMS
- Motherboard : X-S32K2XX-MB (700-31431 REV X3)
- Mini-Module : X-S32K3XXCVB-Q257 (700-47020 REV X2)

2.1 Version Number Details

Version number is of 8 bytes size and can be read at HSE GPR address: 0x4039C020 - - 0x4039C027

FullMem:

00 05 00 00 00 0F 00 06

2.2 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP:

- S32K344
- S32K324
- S32K314

3 Secure-BAF Update

The host can update the Secure-BAF via HSE_SRV_ID_SBAF_UPDATE and service structure "hseSbafUpdateSrv_t" provided by HSE FW in one shot or in streaming mode.

The new Secure-BAF image is programmed at any address in the application Flash memory before calling the Secure-BAF update service.

The new SBAF image is authenticated and decrypted, then stored in the HSE code Flash memory.

On the next reset, the new Secure-BAF boots the HSE Firmware.

Note:

- *NXP recommends updating Secure-BAF in one shot mode.*
- *Secure-BAF update must be performed in a stable environment because any failure in the update process causes the bricking of device.*
- *Secure-BAF update service is only applicable for FULL_MEM configuration.*

Please refer to the **HSE Firmware Reference Manual** for more details.

4 Changes in RTM 0.15.0

Features Added

- ECC handling feature: This feature will prevent the bricking of the device in case reset happens during secure flash area programming..
- Added the feature to make the desired area of application code flash area as OTP. This feature can be used only if corresponding service is provided in HSE Firmware. Currently, this feature is *not supported* by the standard HSE Firmware.
- Added the support of enabling the Firmware feature flag in Boot configuration word (Bit number 9) in addition to UTEST location 0x1B000000. The same bit can enable the firmware installation mode via MU without the need to write the marker value 0xA5 in DCM register. For more details, refer to the **HSE Firmware reference manual**.
- Added the feature in SBAF to support the firmware installation and update in backup feature enable and disable configuration via a bit 10 in Boot configuration word in IVT. The backup disable feature is currently *not supported* in standard HSE firmware.
- Added the feature to disable the option to put application core in recovery mode in case of 8 functional and 8 destructive resets via bit number 8 in Boot configuration word in IVT.

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